D-3597

2023-27

Time : 3 Hours

Maximum Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from all the Parts as directed.

PART–A

(Objective Type Questions)

1. Answer by choosing most appropriate option given for each question : 2×10=20

(a) In test cross a dominant phenotype plant cross with :

(i) A homozygous dominant plant

(ii) A heterozygous plant

(iii) A homozygous recessive plant

(iv) None

(b) Mendel's law of independent assortment is based on the result of :

(i) Test cross

(ii) Back cross

(iii) Monohybrid cross

(iv) Dihydbrid cross

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(c) Cytoplasmic inheritance is also known as :

(i) Mendelian inheritance

(ii) Chromosomal inheritance

(iii) Nuclear inheritance

(iv) Extra-nuclear inheritance

(d) Which of the following chemical mutagen acts as a base analog ?

(i) 5-Bromouracil

(ii) Ethyl methane sulfonate

(iii) Nitrous acid

(iv) Colchicine

(e) UV radiation primarily causes mutations by inducing :

(i) Thymine dimers

(ii) Chromosomal breaks

(iii) Base deletions

(iv) Deamination

(f) The chemical used to induce polyploidy in plants is :

(i) Formaldehyde

(ii) Phenol

(iii) Colchicine

(iv) Mustard gas

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(g) "Puffs" in polytene chromosome represent :

(i) Sites of active DNA replication

(ii) Sites of active RNA synthesis (transcription)

(iii) Area with tightly packed chromatin

(iv) None

(h) A mutation in which purine is substituted by pyrimidine is called :

(i) Transition

(ii) Transversion

(iii) Frameshift

(iv) None

(i) Lampbrush chromosomes are observed during :

(i) Prophase I (Diplotene)

(ii) Prophase I (Leptotene)

(iii) Anaphase I

(iv) Telophase I

(j) A chromosomal aberration in which segment breaks off, rotates by 180 degrees, and reattaches in the reverse direction is called.

(i) Reciprocal translocation

(ii) Inversion

(iii) Duplication

(iv) Deletion

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PART–B

(Short Answer Type Questions)

2. Write short notes on any **four** of following : 5×4=20

(a) Sex-determination

(b) Lampbrush chromosome

(c) Test cross

(d) Chemical mutagens

(e) Cytoplasmic inheritance

(f) Supplementary genes

PART–C

(Long Answer Type Questions)

3. Answer any **three** of the following questions : 10×3=30

(a) Define chromosomal aberration and describe inversion with diagram.

(b) Illustrate cytological basis of crossing over with suitable diagram.

(c) Define mutation and describe molecular basis of gene mutation.

(d) Define interaction of genes and explain complementary gene interaction.

(e) Explain polytene chromosome with suitable diagram.

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